Traffic & Usage Analytics (12 Months)

Date: January 10, 2025 **Report Period:** January 2024 – January 2025

Summary

> This document analyzes traffic and usage patterns for the Signals-API gateway over the past 12 months. The Crypto-AI-Bot has been in active development and paper trading during this period, with the API serving signals to internal monitoring dashboards and early testing clients.

API Request Volume

Total Requests (12 Months)

Aggregate Stats:

- Total Requests: ~4.2M
- Average Requests/Day: ~11,500
- Peak Day: 28,400 requests (Dec 15, 2024)
- Growth Rate: +320% YoY (Jan 2024 vs Jan 2025)

Monthly Request Volume

Request Breakdown by Endpoint

Endpoint	Requests	% of Total
`/v1/signals/health`	2.1M	50%
`/v1/signals/sse` (SSE stream)	1.4M	33%
`/v1/signals/latest`	520K	12%
`/v1/signals/history`	180K	4%
Other (admin, debug)	40K	1%

Key Observations:

- Health checks dominate traffic (automated monitoring every 30s)
- SSE streams account for 33% of requests, representing long-lived connections
- API usage is primarily internal (Grafana, monitoring scripts, testing clients)

User Activity

Active Connections

SSE Stream Connections:

- Average Concurrent Connections: 3-5 (internal dashboards)
- Peak Concurrent Connections: 12 (during load testing)
- Average Session Duration: 45 minutes

Geographic Distribution:

- 100% US-based traffic (Fly.io us-east region, internal testing)
- No external users yet (SaaS launch planned for Q1 2025)

Usage Patterns by Time of Day

Peak Hours (UTC):

- 14:00-22:00 UTC (9am-5pm EST) — Active trading hours
- 60% of traffic occurs during US market hours
- Overnight traffic: Mostly health checks and scheduled tasks

Hourly Traffic Pattern

Growth Trends

Monthly Active Users (Internal)

Definition: Unique IP addresses making authenticated API calls

Month	Active IPs	% Change
Jan 2024	1	—
Apr 2024	2	+100%
Jul 2024	3	+50%
Oct 2024	4	+33%
Jan 2025	5	+25%

Note: All users are internal developers and testing accounts. Public launch pending.

Signal Consumption Rate

Signals Published vs. Consumed:

- Total Signals Published (12 months): ~850K
- Total Signals Consumed via API: ~520K
- Consumption Rate: 61% (remaining signals stored in Redis, pruned after 7 days)

Signal Consumption Rate

Infrastructure Utilization

API Server Load

Average Load (Last 12 Months):

- CPU Utilization: 25-40%
- Memory Usage: 280MB / 512MB (55%)
- Network I/O: 12MB/hour (mostly SSE streams)

Peak Load Events:

- Dec 15, 2024: Load testing (100 concurrent SSE connections)
- Nov 8, 2024: Soak test (48-hour continuous signal publishing)

Redis Bandwidth

Data Transfer (API ↔ Redis):

- Average: 8MB/hour
- Peak: 45MB/hour (during backtesting runs)
- Command Types: 70% `XREAD`, 20% `GET`, 10% `XADD`

API Performance Metrics

Response Times (12-Month Average)

Latency Distribution:

- p50: 22ms
- p75: 45ms
- p95: 95ms
- p99: 240ms

By Endpoint:

- `/health`: 8ms (p50)
- `/latest`: 35ms (p50)
- `/sse`: 12ms (initial handshake), then long-lived

API Latency Over Time

Error Rates

12-Month Aggregate:

- 5xx Errors: 0.03% (120 errors out of 4.2M requests)
- 4xx Errors: 0.18% (mostly 404s from deprecated endpoints)
- Timeout Errors: 0

Root Causes:

- 5xx: Redis connection timeouts during high load (3 incidents)
- 4xx: Misconfigured monitoring scripts hitting old endpoints

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Client Distribution

User Agents

User Agent	Requests	% of Total
Grafana	2.8M	67%
Python Requests	820K	20%
curl	380K	9%
Custom Monitoring	200K	5%

Interpretation:

- Grafana dominates (automated dashboard polling)
- Python scripts are internal monitoring and backtesting tools
- curl usage indicates manual testing and debugging

Referrer Sources

All traffic is internal:

- `localhost` : 15% (local development)
- `fly.io` internal network: 85% (Grafana, monitoring services)

Cost Analysis (Traffic-Related)

Fly.io Bandwidth Costs

12-Month Total:

- Outbound Bandwidth: ~140GB
- Cost: $0 (free tier: 100GB/month per app, not exceeded)

Redis Cloud Costs

12-Month Total:

- Data Transfer: ~95GB
- Cost: $0 (included in managed plan)

Total Infrastructure Cost (Bandwidth): $0

Forecasted Growth (2025)

Projected Traffic (if SaaS launch succeeds)

Assumptions:

- 50 paying subscribers by end of Q1 2025
- 200 subscribers by end of Q4 2025
- Each subscriber polls `/latest` every 5 minutes (avg)

Projected Annual Traffic (2025):

- Requests: ~35M (8x growth from 2024)
- SSE Connections: 50-200 concurrent
- Bandwidth: ~1.2TB/year

Infrastructure Needs:

- Fly.io: Upgrade to 1GB RAM, 2 CPUs (~$30/month)
- Redis: Upgrade to 5GB plan (~$40/month)

Traffic Growth Forecast

Key Takeaways

1. **Stable Growth:** 320% YoY request growth, all internal usage
2. **Low Error Rate:** 0.03% 5xx errors, 99.97% success rate
3. **Scalability Headroom:** Current infrastructure can handle 10x traffic
4. **Zero External Users:** All traffic is internal testing; SaaS launch pending
5. **Cost Efficiency:** $0 bandwidth costs due to low volume

Appendix: Data Sources

- **API Logs:** Fly.io request logs (parsed via custom scripts)
- **Redis Metrics:** Redis Cloud dashboard + Prometheus
- **Grafana:** Custom dashboards tracking `/health` and `/sse` endpoints

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End of Document